

NATIONAL INSTITUTE OF TECHNOLOGY KURUKSHETRA

Department of Mathematics & Computing

Mid Semester Examinations I

MC-A IV Semester

MAPC 204: Computer Oriented Numerical Methods

Date: 21-02-2026

Time: 11.00 AM - 11.50 AM

Max. Marks: 15

Answer the entire question at one place. State all the theorems and results used.

- Q1. Compute the approximate value of $\sqrt[3]{3}$ correct upto three decimal place. Using the Newton Raphson method.
- Q2. The following data defines the sea-level concentration of dissolved oxygen for fresh water as a function of temperature:

t in Celcius	0	8	16	24	32
$s(t)$ in mg/L	14.621	11.843	9.870	8.418	7.305

Estimate the concentration of dissolved oxygen at 27 degrees Celsius.

- Q3. Consider the linear system of equations $x + 2y + 10z = 5$, $10x + 2y + z = 9$, $2x + 10y + z = -7$. Perform three iterations of the Gauss-Jacobi iterative method to obtain an approximate solution of the system.
- Q4. Prove that $\delta = \nabla(1 - \Delta)^{\frac{1}{2}}$ and $\Delta - \nabla = \Delta \nabla$.